

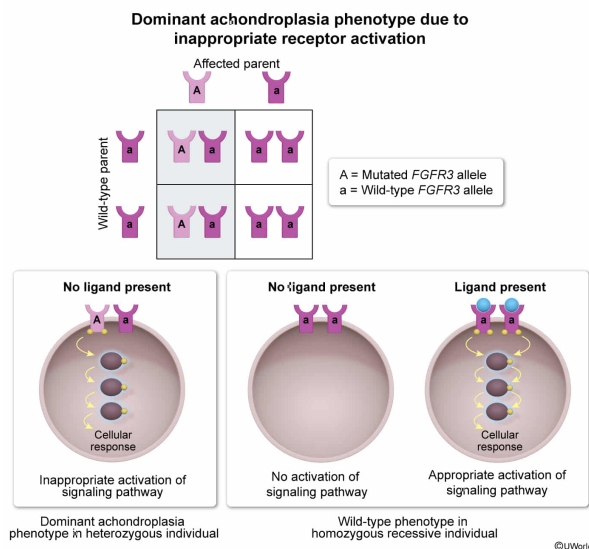
Achondroplasia is a dominantly inherited condition with a phenotype that includes a large head and short stature. It is caused by mutations in the gene for fibroblast growth factor receptor 3 (*FGFR3*), which causes the receptor to be active even when its ligand is not present. Compared to individuals without achondroplasia, individuals who are heterozygous for this condition most likely:

- ☐ A. do not have the achondroplasia phenotype because expression of the wild-type receptor masks the expression of the mutant receptor. [5%]
- ✓ ☒ B. have the achondroplasia phenotype because having one copy of the mutant receptor causes the receptor to be inappropriately activated. [83%]
- ☐ C. have an intermediate phenotype between wild-type and achondroplasia because both wild-type and mutant receptors are expressed. [8%]
- ☐ D. do not have the achondroplasia phenotype because only the wild-type receptor is expressed. [3%]

Correct

83% answered correctly

Explanation:



An individual's **genotype** is the combination of **alleles** for a given trait whereas an individual's **phenotype** refers to the organism's observable characteristics. In the presence of a **dominant** allele, a **recessive** allele is hidden, with no contribution to the individual's phenotype. Therefore, an individual must be **homozygous** for a recessive allele for the mutant phenotype to be observed. This condition is known as **autosomal recessive inheritance**.

However, in the case of **autosomal dominant inheritance**, the mutant phenotype is observed in heterozygous individuals. Because only one allele is required to observe the phenotype, dominant alleles are often the result of gain-of-function mutations, which cause the resulting protein to have new or enhanced activities.

The question states that achondroplasia is a dominantly inherited condition and that individuals with achondroplasia have mutations that cause the FGFR3 receptor to be active even when its ligand is not present. A heterozygous individual would express one wild-type (WT) allele and one mutant allele. Because the **mutation causes receptor activation in the absence of its ligand**, having one mutant allele would lead to **inappropriate activity of the FGFR3 signaling pathway**, and therefore the **achondroplasia phenotype** is observed.

(Choice A) This mutation causes inappropriate receptor activity in the absence of the signal, so expression of the WT receptor *cannot* mask this mutation. In autosomal recessive inheritance, the mutant allele is generally not expressed or is nonfunctional, allowing the dominant allele to mask (or compensate for) the effect of the mutation.

(Choice C) In autosomal dominant inheritance, a full phenotype is seen in heterozygotes when only one mutant allele is present. An intermediate phenotype would be observed in a heterozygote in the case of **incomplete dominance**.

(Choice D) The question states that the mutation causes inappropriate signaling, which indicates that the mutant allele is expressed and *would* be expected to cause a phenotype. In some cases, mutations can cause an allele not to be expressed but this is generally the case in autosomal recessive (*not* dominant) inheritance.

Educational objective:

In autosomal dominant inheritance, a phenotype is observed in the heterozygote when only one mutant allele is present. Mutations that cause new or increased activity of a protein often result in an autosomal dominant inheritance pattern.

Time spent:0 sec

Subject:
Biology

QID:404172

System:
Genetics and Evolution